Food and Chemical Toxicology 50 (2012) 3902–3911

Contents lists available at SciVerse ScienceDirect

Food and Chemical Toxicology

journal homepage: www.elsevier.com/locate/foodchemtox

Ergothioneine protects against neuronal injury induced by β-amyloid in mice

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ARTICLE INFO

Article history:
Received 20 June 2012
Accepted 9 August 2012
Available online 16 August 2012

Keywords:
Ergothioneine
Melatonin
β-Amyloid
Alzheimer's disease
Memory and learning

ABSTRACT

β-Amyloid peptides (Aβ) are neurotoxic and contribute to the development of Alzheimer's disease (AD). Ergothioneine (EGT) has been shown to protect against loss of memory and learning abilities in mice. In this study, mice were orally fed EGT (0.5 or 2 mg/kg body weight) for 16 days before treatment (i.c.v) with a single dose of Aβ1–40 in the hippocampus. After resting for 12 days to restore the body weight, the mice were again fed EGT for additional 39 days. Active avoidance tests were conducted on days 37–39 (short-memory avoidance) and on days 37, 44 and 51 (long-memory avoidance). Water maze task was used to evaluate learning and memory abilities by acquisition test and retention test. In both long-memory avoidance and water maze tests, EGT significantly decreased the escape latency and increased the frequency of successful avoidance. Furthermore, EGT significantly prevented Aβ accumulation in the hippocampus and brain lipid peroxidation, restored acetylcholinesterase (AChE) activity, maintained glutathione/glutathione disulfide ratio and superoxide dismutase activity in brain tissues of Aβ1–40-teated mice. Thus, EGT can protect against Aβ-induced loss of memory and learning abilities in mice. Further studies are required to confirm the protective effects of EGT on the development or progression of AD.

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1. Introduction

Alzheimer's disease (AD) is characterized by progressive neurodegeneration leading to loss of cognitive abilities and ultimately to death. The characteristic neuropathology of AD is the accumulation of beta-amyloid (Aβ) protein to form plaques and tau phosphorylation, resulting in tangle formation of brain regions. Memory deficits and amyloid deposition in brain also have been observed in rats and mice with several different forms of Aβ (Stephan and Phillips, 2005), such as the Aβ1–40, Aβ1–42, and Aβ25–35, Aβ1–56 peptide fragments could influence the learning and memory ability in mice (Lesne et al., 2006). In this respect, many reports have shown that a single intracerebral or intracerebroventricular injection (i.c.v.) of synthetic Aβ peptides could induce major neuropathological signs related to early stages of AD in rodents (Stephan and Phillips, 2005; Klementiev et al., 2007). Strong genetic evidence suggests that it is the aberrant accumulation of Aβ which lies upstream, and Aβ accumulation leads to downstream pathologies such as tangle formation (Oddo et al., 2003), extensive inflammation (Meda et al., 1995), oxidative damage to lipids, proteins and DNA (Hensley et al., 1995), and glycation of proteins (Horie et al., 1997).

Oxidative stress generated from Aβ deposition in the brain of AD patients can result in an increase in free radical generation or an impaired endogenous antioxidant system (Rammouz et al., 2011; Schubert et al., 1995; Townsend and Pratico, 2005). The hallmark modifications of oxidative damage in AD brain include the presence of proteins modified with advanced glycation end products (AGEs), malondialdehyde (MDA), 8-hydroxy-deoxyguanosine (8-OHdG), 4-hydroxynonenal (HNE), and nitrotyrosine, along with increased amounts of lipid peroxidation (Hensley et al., 1996). Thus, many efforts have been directed to find therapeutic agents either synthetic compounds or natural products that could reduce oxidative stress and inflammation to improve the memory. Evidence shows that increased intakes of omega-3 fatty acids (Richter et al., 2010), curcumin (Lin et al., 2011), soy isoflavones (Hsieh et al., 2009), or increased cognitive stimulation (Billings et al., 2007), can protect against formation of AD pathologies.

Ergothioneine (2-mercaptohistidine trimethylbetaine; EGT), which is formed in some bacteria and fungi but not in animals (Melville et al., 1955), has been shown to be radioprotective and to scavenge singlet oxygen, hydroxyl radical, hypochlorous acid and peroxyl radicals as well as to inhibit peroxynitrite-dependent

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0278-6915/$ - see front matter © 2012 Published by Elsevier Ltd.
http://dx.doi.org/10.1016/j.fct.2012.08.021

nitration of proteins and DNA (Aruoma et al., 1997; Dubost et al., 2007). In addition, Jang et al. (2004) have reported that EGT is neuroprotective by protecting rat pheochromocytoma (PC12) cells from oxidative and nitrosative cell death caused by Aβ. In vivo studies have shown that EGT protects retinal neurons against *N*-methyl-D-aspartate-induced excitotoxicity (Moncaster et al., 2002) and diabetic embryopathy in pregnant rats (Guijarro et al., 2002). EGT also maintains cellular homeostasis in neuronal cells challenged with a prooxidant mixture of *N*-acetylcysteine/hydrogen peroxide (Aruoma et al., 1999).

We have recently demonstrated that EGT protects against cisplatin-induced neuronal damage to improve the spatial learning and memory ability of CBA mice (Song et al., 2010), but little is known about the effect of EGT on Aβ-induced learning and memory loss in mice. Thus, the present study was designed to investigate the effect of EGT on learning and memory by active avoidance test and Morris water maze task in C57BL/6 mice. At the end of study, the mice were killed and cerebral cortex and hippocampus were collected for the examination of acetylcholinesterase (AChE) activity and oxidative stress markers: lipid peroxidation, protein damage, GSH/GSSG ratios, and superoxide dismutase (SOD) activity (Li et al., 2011).

2. Materials and methods

2.1. Chemicals

Chemicals including ergothioneine, Aβ1–40, *n*-butanol, thiobarbituric acid, 1,1,3,3-tetraethoxypropane and all other reagents were purchased from Sigma Chemical Company (St. Louis, MO, USA). The beta-amyloid, 17–24 (4G8) monoclonal antibody was exclusively provided by Covance (Catalog number SIG-39220), and that the host species is mouse.

2.2. Animal experimental design

C57BL/6 mice (16 weeks old) weighing 30 ± 2 g were used for the experiments. The mice were maintained on a 12:12 h light: dark cycle and provided with food (LabDiet 5001 Rodent Diet; PMI Nutrition International) and water *ad libitum* throughout the study. EGT and melatonin (as positive control) were both dissolved in normal saline, and Aβ1–40 was prepared in vehicle solvent (0.35% acetonitrile plus 0.1% trifluoroacetic acid (v/v, pH 2.0). Aβ1–40 was administered by a single intracerebroventricular injection (i.c.v.) according to the procedure established by Laursen and Belknap (1986). The dose of Aβ1–40 (400 pmol in 5 μL/mouse) is the same as that reported by Jhoo et al. (2004). The mice were divided into five groups (nine mice per group). The five groups are as follows: group 1 (control): vehicle solvent (5-μL, i.c.v.) + saline, p.o.; group 2 (Aβ 1–40): Aβ 1–40 5 μL, i.c.v.) + saline, p.o.; group 3 (+MEL): Aβ 1–40 (5 μL, i.c.v.) + melatonin, p.o. (10 mg/10 mL/kg bw); group 4 (+L-EGT): Aβ 1–40 (5 μL, i.c.v.) + low EGT, p.o. (0.5 mg/10 mL/kg bw); group 5 (+H-EGT): Aβ 1–40 (5 μL, i.c.v.) + high EGT, p.o. (2 mg/10 mL/kg bw). Mice received EGT or melatonin for 55 days (days 1–16 and days 29–67). On day 17, the mice were treated with a single dose of Aβ (5 μL/mice, i.c.v.) followed by resting for 12 days (days 17–29) because the mice were rather weak after Aβ injection, as evidenced by the loss of body weights. The doses of EGT and melatonin were the same as those used in our previous studies (Song et al., 2010). The high-EGT dose (2.0 mg/kg/day) is equivalent to a human consumption level of 10 mg/day, which can easily be achieved by consumption of EGT-rich foodstuff, such as mushrooms (an intake of approximately 4 g dry mushroom per day would achieve such an intake) (Song et al., 2010).

The experimental scheme is shown in Fig. 1. All experimental procedures involving animals were conducted in accordance with National Institutes of Health (NIH) guidelines. This experiment was approved by the Institutional Animal Care and Use Committee (IACUC) of the Chung Chou University of Science and Technology.

2.3. Infusion of Aβ1–40

Mice model with AD-like phenotype was modified by infusing Aβ1–40 into cerebral ventricle in the brain as described previously (Jhoo et al., 2004; Lee et al., 2007; Yeh et al., 2012). Briefly, mice were anesthetized with isoflurane (flow rate was 10 ml/min). The left skull was exposed and drilled (relative to the bregma; 2.3-mm posterior, ±2.5-mm lateral) according to the atlas of Paxinos and Watson (2005) with a stereotaxic frame (SINGA, Taiwan). The osmotic minipump used in the animal model of AD with impaired memory was filled with Aβ1–40 solution or vehicle solution. The outlet of the infusion needle (gauge 26) was inserted 1.5 mm into the right and left lateral ventricle and attached to the skull with dental cement, and then the minipump was quickly implanted into the backs of the mice. Aβ1–40 solution was infused into the left and right ventricle at 0.5 μL/min for 10 min.

2.4. Active avoidance test

Active avoidance was also measured using a Gemini Avoidance System (San Diego Instrument, San Diego, CA). On the first trial, mice were placed individually into the large compartment of the apparatus with the door closed and accommodated there for about 10 s. A light bulb (60 W) was switched on alternately in the two compartments for 10 s and used as a conditioned stimulus (CS). The CS preceded the onset of the unconditioned stimulus (US) by 5 s. The US was an electric shock (0.3 mA for 5 s) applied to the floor grid. If the animal avoided the US by running into the dark compartment within 5 s after the onset of the CS, the microprocessor recorder unit of the shuttle-box would record an avoidance response. Each mouse was given five trials daily for 3 days with a mixed inter-trial interval of 20 s. The activity level was also assessed by measuring the number of crossings between the chambers when no shock was present (intertrial crossing). The recorder unit of automated shuttle-box continuously recorded this parameter during all experimental periods (15 trials). The results are expressed as the mean percentage avoidance responses for each daily shuttle-box session. The more avoidance number and less latency time the mouse escapes, the better learning memory the mouse has. The active avoidance tests that were conducted on days 37, 44 and 51 (with a 7-day interval) are referred to as the long-memory avoidance tests; while those performed for three consecutive days (days 37 through 39) are referred to as the short-memory avoidance tests.

2.5. Water maze task

The Morris water maze task with some modification was used to evaluate the memory and learning abilities that are referred to as the acquisition test (conducted from days 58 to 60 and on day 67) and the retention test (conducted on day 67). A stainless steel circular tank (diameter 160 cm, height 60 cm) was used as the apparatus of the water maze in which a movable escape platform (diameter 9 cm, height 26 cm) was hidden inside the tank. The tank was filled to a height of 28 cm with water of approximately 23 ± 2 °C that was clouded with powdered milk. The circular tank was divided into four quadrants (I, II, III, and IV), and a position with equal distance from center and edge in the middle of each quadrant was marked for the location of platform. The water tank was located in a test room with many cues external to the maze. The room had adjustable indirect light, and a camera was set the ceiling above the center of water tank, and the record program was the SINGA Real-Time Trace Mouse Ver. 1.17 System (Taipei, Taiwan). The position of the cues remained unchanged throughout the water maze task.

2.5.1. Water maze acquisition test

Acquisition test was conducted four times a day on days 58–60 and on day 67 after EGT administration. A transparent platform was set inside the tank, and its top was submerged 1 cm below the water surface in the center of one of the four quadrants of the maze. For each trial, the mouse was put into the pool at one of the four quadrants, the sequence of the positions being selected randomly. The platform was hidden in a constant position throughout the test period in the middle of one quadrant, equidistant from the center and the edge of the pool. In each trial, the latency to escape onto the hidden platform was recorded. If the mouse found the platform, it was allowed to remain there for 30 s and was then returned to its home cage. If the mouse was unable to find the platform within 180 s, the training was terminated and a maximum score of 180 s was assigned. The successful crossing yield (%) of mice in the acquisition test on days 58–60 and day 67 was calculated as follow: [the successful number crossing to the hidden platform/(the successful number + defeat number crossing to the hidden platform] × 100.

2.5.2. Water maze retention test

Retention test, which was conducted on day 67 after EGT administration, is different from acquisition test in that the hidden platform was removed from the pool in the retention test. Each mouse was allowed to swim for 60 s in the water maze and the test was carried out once. The route for each mouse to cross the annulus (where the hidden platform was located in the water maze acquisition test) was recorded.

2.6. Preparation of brain tissue homogenates

After completion of treatment with EGT (on day 68), the mice were killed by decapitation under CO_2 anesthesia. All following procedures were carried out at 0–4 °C. Whole brains, except cerebellum, were homogenized using an Ultra Turax homogenizer in 1:10 (w/v) buffer (5.0 mM Tris base, 150 mM NaCl and 20 mM EDTA, pH 7.5). The homogenates were sonicated for 30 s in a disruptor (Bronson Sonic, NY) and centrifuged at 16,000*g* for 10 min. The supernatant was kept at −70 °C until use.

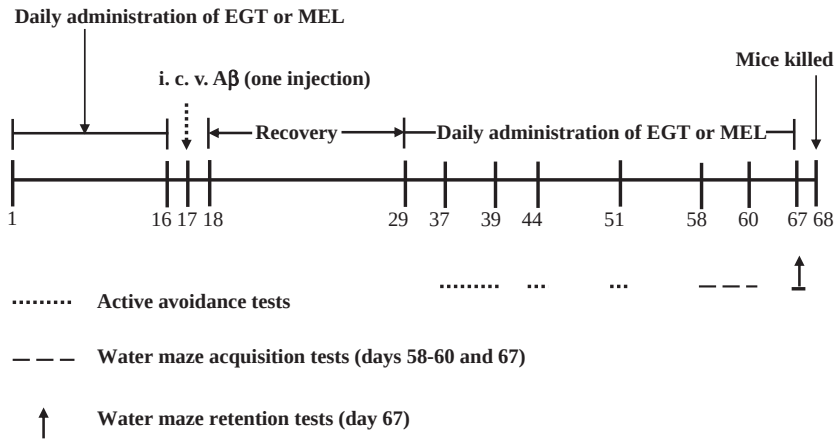


Fig. 1. Experimental scheme for the treatment of Aβ and ergothioneine (EGT) or melatonin (MEL) as well as the active avoidance tests and water maze tests.

2.7. Determination of GSH/GSSG ratio in brain tissues

GSH/GSSG ratios in brain tissue homogenates were assayed according to the HPLC method of Schofield and Chen (1995) with some modification. In brief, after adding100 μL of 10% perchloric acid to precipitate the protein of brain homogenates (1.0 mL), then the supernatants were treated with iodoacetic acid in the dark at 40 °C for 1 h. A volume (0.2 mL) of 3% (v/v) 2,4-dinitrofluorobenzene was added to the reaction mixture and allowed to react at 40 °C for 4 h in the dark. After centrifugation (3000*g* for 15 min), a portion (25 μL) of the supernatant was applied onto HPLC column. The HPLC system consisted of a Hitachi l-6200 pump, a Hitachi D-2500 chromatointegrator, a Hitachi L-4200 UV–vis detector, and a 3-aminopropyl column (200 × 4 mm, 5 μm). The mobile phase consisted of (A) water/methanol (1:4, v/v) and (B) 2 M sodium acetate, pH 4.6/methanol (36:64, v/v). The 3-aminopropyl column was eluted by a 30 min gradient from 75% (A) to 5% and then isocratically maintained for 15 min at a flow rate of 1.2 mL/min and UV detection at 365 nm. The GSH/GSSG ratio was determined as GSH content/GSSG content.

2.8. Determination of thiobarbituric acid-reactive substances (TBARS) in brain tissues

Lipid peroxidation in brain tissue homogenates was determined using the thiobarbituric acid (TBA) assay (Buege and Aust, 1978). Butylated hydroxytoluene (10 μL, 50 mM) was added to the tissue homogenate (1.0 mL) to terminate the peroxidation reaction and then mixed with 1 mL of 7.5% (w/v) cold trichloroacetic acid (TCA) to precipitate proteins. The supernatant was allowed to react with 1 mL of 0.8% (w/v) TBA in a boiling water bath for 45 min. After cooling, levels of TBARS were determined at 555 nm with excitation at 515 nm using 1,1,3,3-tetraethoxypropane as the standard and were expressed as nmol per mg of protein.

2.9. Determination of EGT contents in brain tissues

For determination of EGT contents, we used the method outlined in detail by Dubost et al. (2007) for measuring EGT in mushrooms. In short, EGT was measured using HPLC with separation carried out on two Econosphere C18 columns (Thermo scientific Associates, IL) with each column being 250 mm × 4.6 mm, 5 μm particle size connected in tandem. The isocratic mobile phase was 50 mM sodium phosphate in water with 3% acetonitrile and 0.1% triethylamine adjusted to a pH of 7.3 with a flow rate of 1 ml per minute. The injection volume was 20 μL, with the column temperature being ambient. EGT was quantified by monitoring absorbance at 254 nm using the authentic standard. Data were expressed as mmol of EGT per milligram of protein (mmol/mg protein). Protein concentrations were determined using a standard commercial kit (Bio-Rad Laboratories Ltd.) with serum bovine albumin as standard.

2.10. Determination of superoxide dismutase (SOD) and acetylcholinesterase (AChE) activities

SOD activity in brain tissues was assayed using the pyrogallol autoxidation method (Nandi and Chatterjee, 1988). AChE activity in brain tissues was measured by the spectrophotometric method as developed by Ellman et al. (1961).

2.11. Immunohistochemical stain of Aβ in hippocampus

The brain was fixed in 10% formalin at pH 7.4. Brains tissues were blocked, and serial 2.5-μm-thick frozen sections cut on a sledge microtome were collected sequentially and without interruption into wells. The tissue was dried at 60 °C for 1 h, then deparaffinized and hydrated, following a dip of xylene, gradient concentration of ethanol, and PBS. These sections were treated for 30 min in 3% hydrogen peroxide in absolute methanol, followed by several washes with cold PBS. After blocking in 10% normal goat serum in PBS, sections from regions containing hippocampus were processed for Aβ immunohistochemical stain by placing in primary monoclonal Aβ1–40 antibody (1:100) diluted in antibody diluents at 4 °C overnight. Labeling was detected by using the nonbiotin polyhydrogen peroxidase immunohistochemistry detection kit (BioGenex, San Ramon, CA). Counterstaining of the section was then carried out with hematoxylin for 3 min and stopped with a PBS wash. Sections were then mounted, air dried, and coverslipped. The Aβ1–40 accumulation in hippocampus was monitored by intermittent microscope examination and digital image system (Image Pro Plus 6.2) provided by Total Smart Technology Co., Ltd. (Taipei, Taiwan). The percentage of Aβ accumulation was expressed as: total percentage (total Aβ plaque area in cover slipped)/(total cortex area in cover slipped) × 100%.

2.12. Statistical analysis

Data are expressed as means ± SD and analyzed using one-way ANOVA followed by LSD Test for multiple comparisons of group means. When the *F* values of ANOVA were significant, the group means were then compared using the Lease Significant Differences (LSD) test. All statistical analyses were performed using SPSS for Windows, version 10 (SPSS, Inc.); a P value <0.05 is considered statistically significant.

3. Results

3.1. Effect of EGT on body weights of mice treated with Aβ

As shown in Fig. 2, Aβ1–40 alone significantly decreased body weights of C57BL/6 mice after Aβ1–40 treatment (2–4 weeks), as compared to those of control mice. However, supplementation with EGT (0.5 or 2 mg/kg bw) or melatonin (10 mg/kg bw) resulted in substantial recovery of body weights. In addition, only the body weight of H-EGT group was significantly higher than that of Aβ control, and there was no statistical difference in the mean body weights between L-EGT and H-EGT groups.

3.2. Active avoidance test

The active avoidance test, i.e., the escape latency and number of avoidance, was used to evaluate the memory and learning ability in this animal model. Active avoidance tests were performed using the shuttle-box active avoidance test and presented as both short-memory retention (conducted for three consecutive days on days 37 through 39) and long-memory retention (conducted on days 37, 44, and 51; i.e., with a 7-day interval). As shown in Fig. 3, all groups of mice in the short-memory avoidance tests showed gradual decreases in the escape latency (i.e., they could escape to the other chamber before the electronic shock) (Fig. 3A) and increases in the number of avoidance (Fig. 3B) on the 2nd day (day 38) and the 3rd day (day 39), as compared to those of the 1st day (day 37).

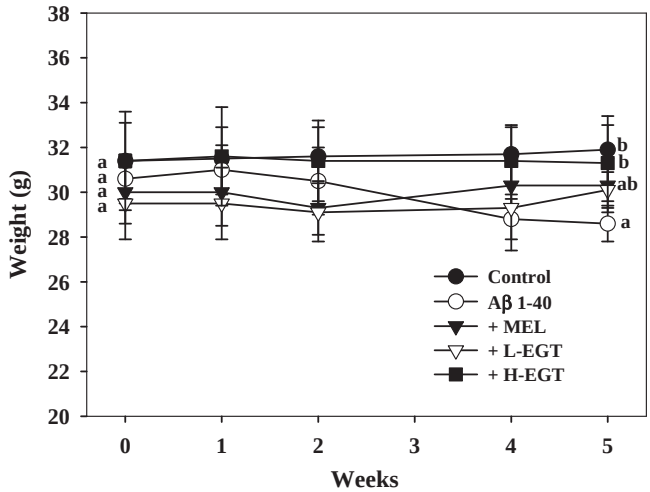


Fig. 2. Effects of ergothioneine (EGT) and melatonin (MEL) on body weight changes in mice treated with Aβ1–40. Two groups of mice were infused i.c.v. with either saline solution (control group) or Aβ1–40 solution (Aβ group) without administration of test materials. The other mice infused with Aβ1–40 were orally administered MEL (10 mg/kg/d as positive control), low-dose EGT (+L-EGT, 0.5 mg/kg/d) or high-dose EGT (+H-EGT, 2 mg/kg/d). Values (means ± SD, $n = 9$ mice per group) obtained on the same week not sharing an alphabetic letter are significantly different ($P < 0.05$).

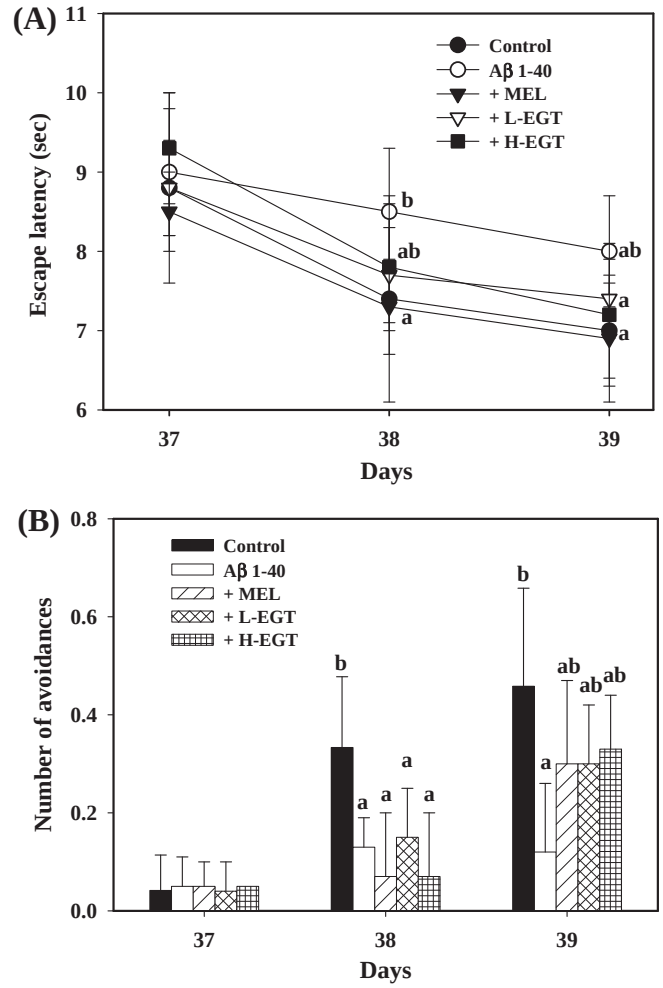


Fig. 3. Effects of ergothioneine (EGT) and melatonin (MEL) on active-avoidance test (short-memory test) in mice treated with Aβ1–40. Two groups of mice were infused i.c.v. with either saline solution (control group) or Aβ1–40 solution (Aβ group) without administration of test materials. The other mice were infused with Aβ1–40 and orally administered MEL (10 mg/kg/d as positive control), low-dose EGT (+L-EGT, 0.5 mg/kg/d) and high-dose EGT (+H-EGT, 2 mg/kg/d). The tests were carried out on days 37–39 to record the escape latency (A) and number of avoidance (B). Values (means ± SD, $n = 9$ mice per group) obtained on the same treatment day not sharing an alphabetic letter are significantly different ($P < 0.05$).

Although the escape latency and number of avoidance on day 39 were improved in mice treated with low-dose or high-dose EGT, as compared to those of Aβ1–40 treatment alone, the differences were not statistically significant due to relatively large variations of the results. In contrast, melatonin significantly decreased the escape latency (7.3 s), as compared to those of Aβ1–40-treated mice (8.7 s) on the second day (Fig. 3A), although no significant changes were observed in the number of avoidance (Fig. 3B).

Long-memory avoidance tests (Fig. 4A) show that the escape latency in the control mice gradually decreased on days 44 and 51, possibly because the mice had learned and memorized from the first experience conducted on day 37. In contrast, the mice treated with Aβ1–40 alone showed significantly increased escape latency on days 44 and 51 ($P < 0.05$), as compare to the control group. Supplementation with both doses of EGT decreased the escape latency to some extent on day 44, and the effects were significant ($P < 0.05$) on day 51, as compared with Aβ treatment alone. Melatonin treatment tended to decrease the escape latency on days 44 and 51, but the effects were not statistically significant. In contrast to the escape latency, the mean number of avoidance in mice treated with Aβ1–40 alone was significantly lower than the control group ($P < 0.05$) (Fig. 4B). However, treatment with EGT or melatonin significantly increased the number of avoidance, as compared to Aβ1–40 treatment alone ($P < 0.05$), although there was no significant difference between the low-dose EGT (0.5 mg/kg) and high-dose EGT (2 mg/kg) groups ($P > 0.05$).

3.3. Water maze task

3.3.1. Water maze acquisition test

Acquisition test in a water maze was performed on days 58–60 and 67 by recording the escape latency and counting the number of crossings over the platform position. As shown in Fig. 5A, the escape latency gradually decreased for each group (even in Aβ treatment group) with each testing day, possibly because these mice had become more familiar with the test. However, even with this improvement, the differences between the Aβ control group and the other groups became more pronounced on latter testing days. This was evidenced by the observation that the escape latency for the Aβ control mice was significantly higher than the other groups only on days 60 and 67, but not on days 58 and 59. In contrast, supplementation with MEL or EGT at either low or high dose significantly decreased escape latency, as compared with Aβ1–40 treatment alone on days 60 and 67 (Fig. 5A) ($P < 0.05$). The successful crossing yield (%) of Aβ1–40-treated mice in the acquisition test on days 60 and 67 was significantly lower than that of the control group (Fig. 5B). The yield of successful crossings was significantly higher in mice receiving low- or high-dose EGT than in mice treated with Aβ1–40 alone on days 60 and 67 ($P < 0.05$) (+MEL group only on day 67) (Fig. 5B).

3.3.2. Water maze retention test

The results of water-maze retention tests clearly demonstrate that mice treated with Aβ1–40 alone searched the target platform with directionless escape and swam around whole pools (Fig. 5C). In contrast, mice supplemented with MEL, low-dose or high-dose EGT swam near the target quadrant, and their swimming routes were significantly decreased, as compared with mice treated with

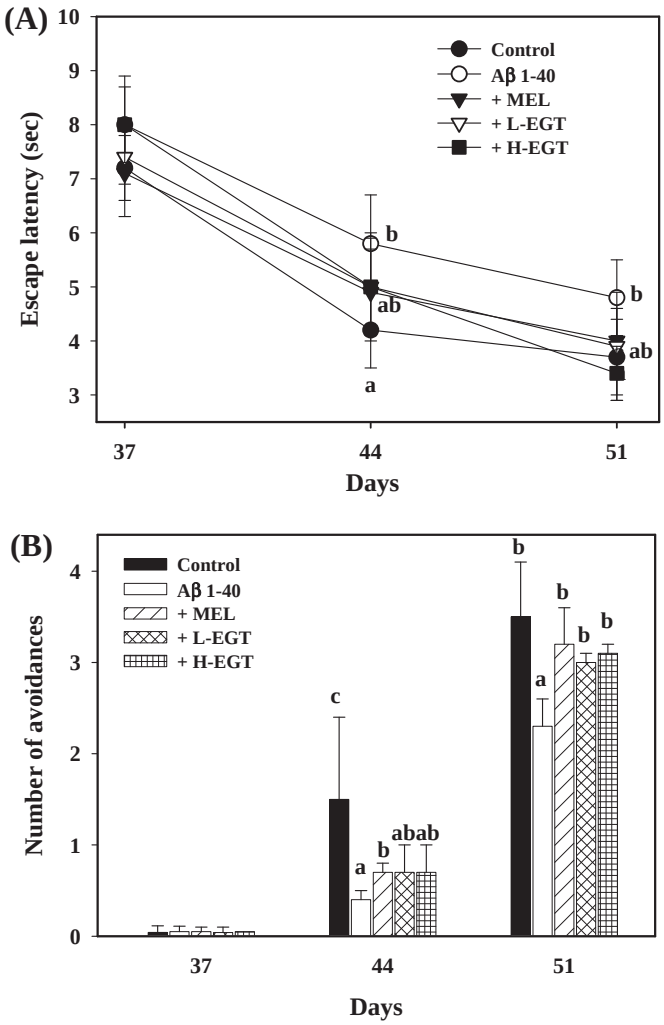


Fig. 4. Effects of ergothioneine (EGT) and melatonin (MEL) on active-avoidance test (long memory test) in mice treated with Aβ1–40. Two groups of mice were infused i.c.v. with either saline solution (control group) or Aβ1–40 solution (Aβ group) without administration of test materials. The other mice were infused with Aβ1–40 and administered MEL (10 mg/kg/d as positive control), low-dose EGT (+L-EGT, 0.5 mg/kg/d) and high-dose EGT (+H-EGT, 2 mg/kg/d). The tests were carried out on days 37, 44 and 51 to record the escape latency (A) and number of avoidance (B). Values (means ± SD, $n = 9$ mice per group) obtained on the same treatment day not sharing an alphabetic letter are significantly different ($P < 0.05$).

Aβ1–40 alone. The distance to hidden platform within 60 s of the Aβ control group was significant higher than that of other treatment groups ($P < 0.05$) (Fig. 5D). The results indicate that MEL and EGT ameliorated the impairment of memory and learning ability of mice induced by Aβ1–40.

3.4. Effect of EGT on Aβ accumulation in hippocampus of Aβ-induced mice

Fig. 6A shows that Aβ accumulation in hippocampus of the Aβ group was higher than that in the control group. Importantly, mice from the MEL group, the low-EGT and the high-EGT groups had less Aβ accumulation in hippocampus than those of the Aβ group. Quantification of the accumulation of Aβ plaques in the hippocampus of mice reveals that the number of Aβ plaque (60) and the percentage of Aβ (0.25%) in total cortex area in Aβ-infused mice were significantly higher than those of control group (27 and 0.09%, respectively, $P < 0.05$) (Fig. 6B). The administration of low-dose EGT and high-dose EGT significantly decreased Aβ plaque formation from 60 to 35 and 31, respectively, and decreased the yield of Aβ plaque area from 0.25% to 0.12% and 0.10%, respectively ($P < 0.05$). There was no significant difference between the low-dose EGT group and the high-dose EGT group ($P > 0.05$) in Aβ plaque formation. MEL treatment did not significantly reduce the number of Aβ plaques, whereas this hormone significantly reduced the percentage of Aβ plaque in total area of the brain (from 0.25% to 0.15%, $P < 0.05$).

3.5. Effect of EGT on AChE activity in the brain of Aβ -induced mice

In mice treated with Aβ1–40 alone, the brain AChE activity was significantly increased in comparison with the control group (Fig. 7). Oral administration with low-dose EGT or high-dose EGT or MEL significantly prevented the rise in brain AChE activity induced by Aβ ($P < 0.05$). However, there was no significant difference in the brain AChE activity between the high-dose EGT group and the low-dose EGT group.

3.6. Effect of EGT on antioxidant status in the brain of Aβ-induced mice

As shown in Table 1, the EGT contents in the brain tissues of mice treated with Aβ1–40 alone (0.55 ± 0.20 mmol/mg protein) were lower (by 38%) than those of control mice (0.34 ± 0.16 mmol/mg protein), although the difference was not statistically significant ($P > 0.05$). Oral administration of low-dose EGT restored the EGT levels in brain tissues, and the brain EGT contents in the high-dose EGT group were significantly higher than those of the control group ($P < 0.05$). In contrast, oral MEL treatment did not affect the loss of EGT induced by Aβ in brain tissues.

As also shown in Table 1, treatment with Aβ1–40 alone significantly increased lipid peroxidation in the brain, as evidenced by a 1.3-fold increase in TBARS. Supplementation with EGT (either at 0.5 or 2 mg/kg body weight) or melatonin completely inhibited Aβ-induced lipid peroxidation. In contrast, Aβ1–40 treatment alone markedly decreased the GSH/GSSG ratio (from 11.8 ± 1.4 to 4.4 ± 1.9). However, treatment with EGT or melatonin significantly reduced the decline of the GSH/GSSG ratio induced by Aβ, and the effect of EGT appeared to be dose-dependent. Similarly, Aβ1–40 treatment alone markedly decreased brain SOD activity (from 6.1 ± 0.6 unit/mg protein/min in the control group to 3.1 ± 0.6 unit/mg protein/min), whereas low-dose EGT treatment was already sufficient to restore the SOD activity. Melatonin treatment only partially restored the brain SOD activity ($P > 0.05$).

4. Discussion

The sulfur-containing EGT is regarded as an antioxidant due to its ability to reduce the oxidative stress both in vitro (Jang et al., 2004; Aruoma et al., 1999) and in vivo (Guijarro et al., 2002). Although several thiols have been reported to protect against Aβ toxicity (Kosuge et al., 2003; Pensalfini et al., 2008; West et al., 2008), it is unclear whether EGT is neuroprotective against Aβ toxicity. The present results show that EGT significantly improved the long-term learning and memory deficits by decreasing the escape latency and increasing the successful number of escape by the electric shock in mice treated with Aβ. In the short-memory avoidance tests (conducted for three consecutive days on days 37–39), EGT also tended to improve the escape latency (on days 38 and 39) and the number of avoidance (on day 39), as compared with Aβ1–40 treatment alone, although the effects were not statistically significant. In addition, EGT significantly improved learning and memory deficits in the water maze acquisition test by decreasing the escape latency and increasing the successful crossing yield to the platform (on days 60 and 67) in mice treated with Aβ alone.

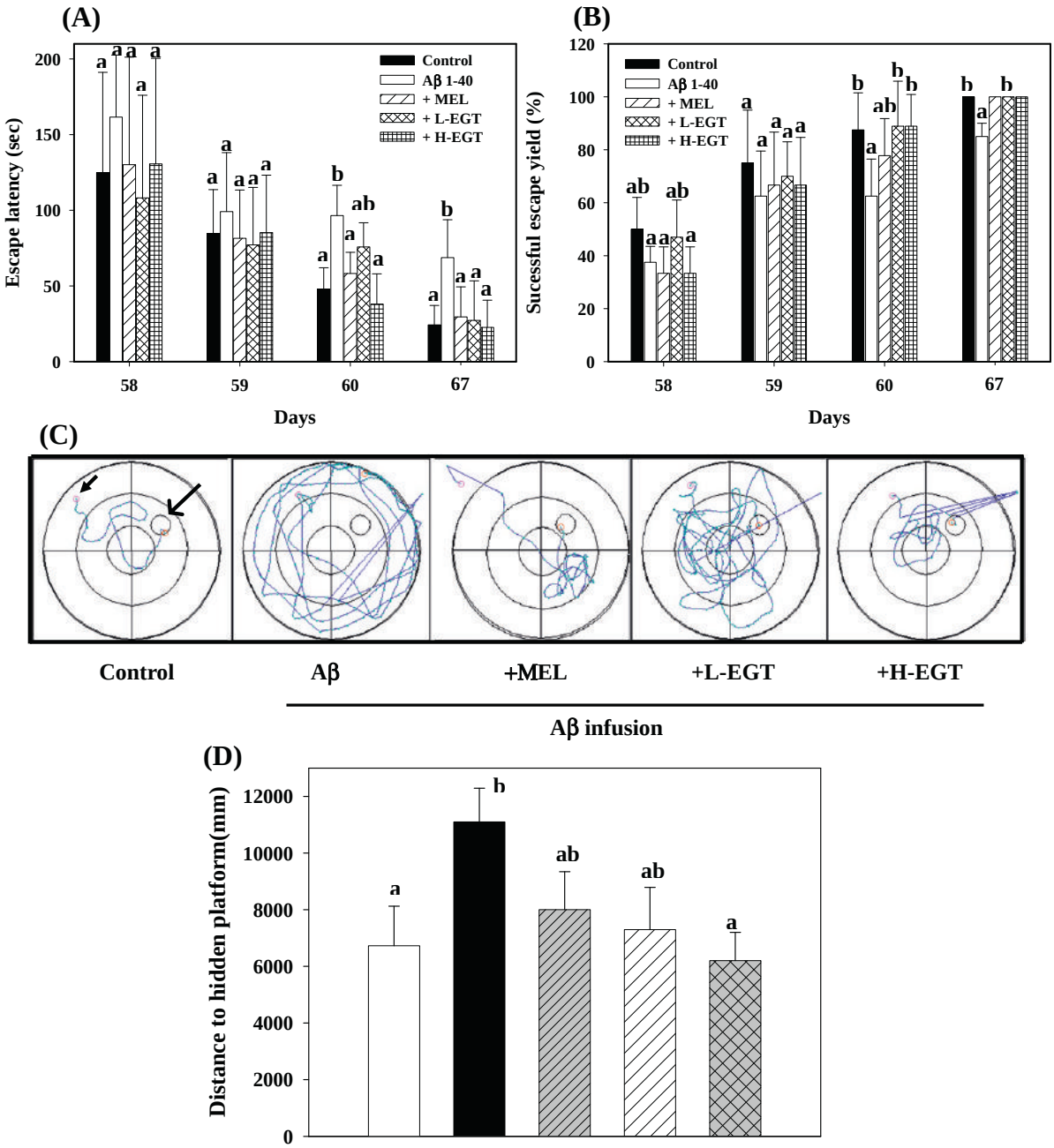
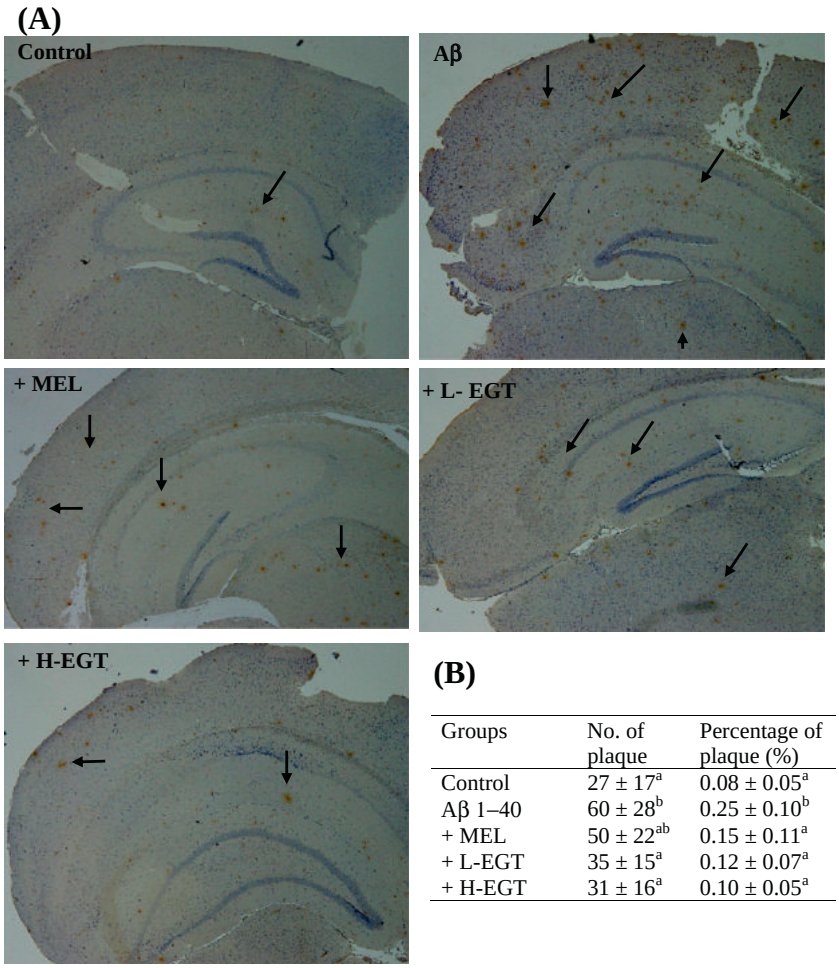


Fig. 5. Effects of ergothioneine (EGT) and melatonin (MEL) on learning and memory deficits of mice treated with Aβ1–40, as evaluated by water maze acquisition tests and retention tests. Two groups of mice were infused i.c.v. with either saline solution (control group) or Aβ1–40 solution (Aβ group) without administration of test materials. The other mice were infused with Aβ1–40 and administered MEL (10 mg/kg/d as positive control), low-dose EGT (+L-EGT, 0.5 mg/kg/d) and high-dose EGT (+H-EGT, 2 mg/kg/d). In acquisition tests, the escape latency (seconds) (A) and the successful crossing yield (%) to the hidden platform were measured on days 58–60 and 67 (B). In retention tests, the swimming routes (C) and the distance of hidden platform (D) of mice were recorded on day 67. The small arrowhead points the initial place where the mouse was put into the pool. The large arrowhead points the location of hidden platform (removed in retention tests). Data (means ± SD, $n = 9$ mice per group) obtained on the same treatment day not sharing an alphabetic letter are significantly different ($P < 0.05$).

EGT supplementation also improved the learning and memory in the water maze retention tests (on day 67), such that the mice swam directionally toward the quadrant where the hidden platform was located. EGT also significantly prevented brain lipid peroxidation, restored brain GSH/GSSG ratios and SOD activity, and attenuated the rise of AChE activity induced by Aβ treatment. To our knowledge, the present study is the first demonstration that pretreatment with EGT prevents Aβ-induced memory and learning impairment in mice.

Aβ is a major risk factor for memory deficit and dementia in AD patients (Schubert et al., 1995; Miklossy, 2011) and is directly responsible for the production of reactive oxygen species (ROS). According to the Aβ theory, deposition of the aggregated Aβ becomes the initiating factor for multiple neurotoxic pathways, which may include excitotoxicity, Ca^{+2} homeostatic disruption, ROS production, GSH depletion, and neuro-inflammation (Gahtan and Overmier, 1999; Ray et al., 2011). The accumulation of excessive ROS leads to lipid peroxidation, protein oxidation, enzyme inactivation, oxidative DNA damage, all of which can lead to cell injury and death (Allen and Tresini, 2000). Therefore, many researchers have tried to find natural antioxidants and anti-inflammatory agents to reduce brain oxidative damage and to ameliorate learn-



(B)

Groups	No. of plaque	Percentage of plaque (%)
Control	27 ± 17^{a}	0.08 ± 0.05^{a}
Aβ 1–40	60 ± 28^{b}	0.25 ± 0.10^{b}
+ MEL	50 ± 22^{ab}	0.15 ± 0.11^{a}
+ L-EGT	35 ± 15^{a}	0.12 ± 0.07^{a}
+ H-EGT	31 ± 16^{a}	0.10 ± 0.05^{a}

Fig. 6. Effects of ergothioneine (EGT) and melatonin (MEL) on the Aβ accumulation in the hippocampus (indicated with arrows) of Aβ1–40-infused mice. Two groups of mice were infused i.c.v. with saline solution (control group) or Aβ1–40 solution (400 pmol) (Aβ group) without administration of test materials. The other mice were infused with Aβ1–40 and orally administered MEL (10 mg/kg/d as positive control), low-dose EGT (+L-EGT; 0.5 mg/kg/d) and high-dose EGT (+H-EGT, 2 mg/kg/d). (A) Aβ accumulation in hippocampus stained by brown dye (arrowhead indicated); the nucleus was stained by blue dye. (B) Quantification of Aβ plaques in the hippocampus by microscope automation and digital image system (Image Pro Plus 6.2). Values (means ± SD, $n = 9$ per group) not sharing an alphabetic letter are significantly different ($P < 0.05$).

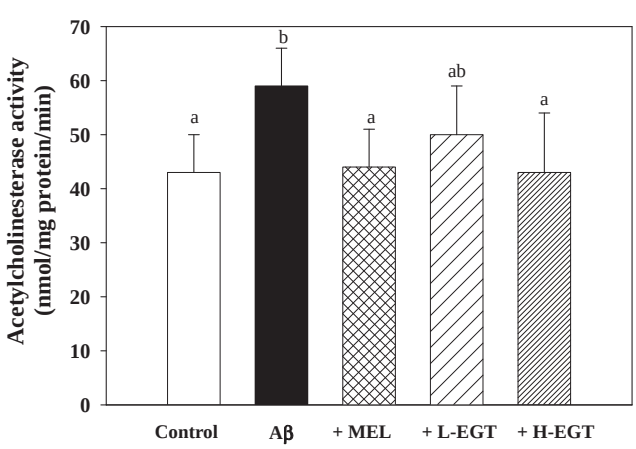


Fig. 7. Effects of ergothioneine (EGT) and melatonin (MEL) on the acetylcholinesterase activity in brain tissues of mice treated i.c.v. with Aβ1–40. Two groups of mice were infused i.c.v. with saline solution (control group) or Aβ1–40 solution (400 pmol) (Aβ group) without administration of test materials. The other mice were infused with Aβ1–40 and orally administered MEL (10 mg/kg/d as positive control), low-dose EGT (+L-EGT; 0.5 mg/kg/d) and high-dose EGT (+H-EGT, 2 mg/kg/d). Values (means ± SD, $n = 9$ mice per group) not sharing an alphabetic letter are significantly different ($P < 0.05$).

ing and memory deficits (Lee et al., 2007; Schubert et al., 1995). However, clinical trials are often unsuccessful because many of the compounds tested have difficulty in crossing the blood–brain barrier (BBB) in the adult (Halliwell, 2001).

It has been shown that endogenous thiols such as metallothionein and GSH are neuroprotective in experimental models of AD (Pensalfini et al., 2008; Aksenov and Markesbery, 2001). GSH is one of the most abundant intracellular nonprotein thiols in the central nervous system, where it plays a major antioxidant role within both neurons and nonneuronal cells (Schulz et al., 2000). However, GSH is poorly transported into most cells or tissues. For this reason many attempts have been made to develop GSH derivatives that are able to easily cross the membranes of many cell types, and to mimic GSH antioxidant properties. For instance, endogenous GSH levels can be increased by dietary or pharmacological intake of GSH precursors or GSH mimics (Kosuge et al., 2003; Lee et al., 2007; Butterfield et al., 2002). Previously, Deiana et al. (2004) have reported that supplementation with EGT not only protects the organs against lipid peroxidation but conserves the consumption of endogenous GSH and alpha-tocopherol.

Our present results (Fig. 6) show that Aβ-infused mice with memory deficit had substantial amounts of Aβ accumulation in the cortex and hippocampus. It should be pointed out that the Aβ 17–24 (4G8) monoclonal antibody used in our study to detect

Table 1
Effects of ergothioneine (EGT) and melatonin (MEL) on the antioxidant status in brain tissues of mice treated i.c.v. with Aβ1–40. The Aβ1–40-infused mice were supplemented with MEL (10 mg/kg/d; as positive control), low-dose EGT (0.5 mg/kg/day; + L-EGT) or high-dose EGT (2 mg/kg/day; +H-EGT)[b].

Groups	EGT (mmol/mg protein)	TBARS[a] (nmol/mg protein)	GSH/GSSG ratio	SOD (unit/mg protein/min)
Control	0.55 ± 0.20ab	4.7 ± 2.3a	11.8 ± 1.4c	6.1 ± 0.6a
Aβ 1–40	0.34 ± 0.16a	6.1 ± 1.4b	4.4 ± 1.9a	3.1 ± 0.6b
+MEL	0.38 ± 0.20a	4.8 ± 0.8a	6.4 ± 1.6b	4.8 ± 1.3ab
+L-EGT	0.62 ± 0.10b	4.6 ± 0.4a	5.1 ± 1.8ab	6.0 ± 1.3a
+H-EGT	0.82 ± 0.21b	4.4 ± 1.0a	7.1 ± 2.3b	6.4 ± 1.3a

[a] TBARS: thiobarbituric acid-reactive substances; GSH/GSSG: reduced glutathione/oxidized glutathione; SOD: superoxide dismutase.
[b] Values (means ± SD, $n = 9$ for the test groups) not followed by a common alphabetic letter are significantly different ($P < 0.05$).

Aβ deposition is sensitive for such purposes in both mice and humans, as this antibody has been used to reduce cerebral amyloid angiopathy and associated micro-hemorrhages in aged mice (Thakker et al., 2009). The Aβ accumulation in the cortex and hippocampus of Aβ-infused mice was significantly suppressed by EGT administration. Importantly, Gründemann et al. (2005) have reported that EGT has an organic cation transporter-1 (OCTN1) in the neuronal cells that can pass the BBB to enter the central nervous system by maintain normal membrane fluidity thereby reducing neuronal cell apoptosis. It is unclear why the control mice fed the basal diets have relatively high levels of EGT, but it is possible that the diet may contain some amounts of EGT. This could lead to accumulation of EGT in the control mice because of the highly specific OCTN1 for the uptake of EGT in both humans and animals (Gründemann, 2012). Herein, we showed that the EGT levels were dose-dependently increased in brain tissues of mice supplemented with EGT, suggesting that EGT can enter brain tissues.

Cognitive deficits associated with Alzheimer's disease (AD) are thought to be primarily related to the degeneration of cholinergic neurons in cerebral cortex and hippocampus, resulting in deficits of cholinergic neurotransmission. AChE is responsible for acetylcholine hydrolysis, from both cholinergic and non-cholinergic neurons of the brain (Atack et al., 1983), whose activity has been shown to be increased within and around amyloid plaques (Ulrich et al., 1990; Morán et al., 1993), to promote the assembly of Aβ-peptides into fibrils (Inestrosa et al., 1996) and to increase the cytotoxicity of these peptides (Alvarez et al., 1998). Approaches to improve cognitive and behavior symptoms of AD have targeted the inhibition of AChE to increase concentrations of ACh (Krall et al., 1999). As indicated by Kaizer et al. (2005), the elevation of AChE activity may be due to a direct neurotoxic effect on the plasma membrane caused by increased lipid peroxidation, and such changes in plasma membranes may affect the integrity and functionality of the cholinergic system. Thus, alterations in the lipid membrane could be a decisive factor in changing the conformational state of the AChE molecule, which would result in learning and memory deficits in mice upon exposure to Aβ1–40. Our present observation that EGT was capable of preventing the increase of AChE activity suggests that EGT may play an important role in the maintenance of acetylcholine synaptic levels by preventing the rise of AChE activity. Thus, the prevention or the restoration of brain AChE activity may be another mechanistic action by which EGT improves learning and memory functions of mice.

Although little is known about the safety of EGT consumption, no animal studies have shown any deleterious effects so far, including our present mouse study. It should be noted that an early study showed that some diabetic patients had elevated blood EGT levels (Fraser, 1950), and this had led to the speculation that EGT may induce diabetes through chelating of zinc (Epand, 1982). However, later experiments have found no statistical significance in blood EGT levels between diabetics and non-diabetics (Epand and Epand, 1988).

MEL (*N*-acetyl-5-methoxytryptamine), which was used as a positive control in the present study, is the main hormone of the pineal gland that has been described to be a potent antioxidant (Tan et al., 1993, 2003). Studies show that melatonin levels are lower in AD patients compared with those in age-matched control subjects (Liu et al., 1999; Ozcankaya and Delibas, 2002). It is generally accepted that a MEL deficit is closely related to aging and age-related diseases (Wu and Swaab, 2005). Indeed, MEL can prevent toxic aggregation of Aβ peptide, and when taken long term, MEL can protect against cognitive deficits in APP transgenic mice (He et al., 2010). In addition, MEL has been reported to either ameliorate or prevent the toxicity of cisplatin (Song et al., 2010; Hara et al., 2001). Agrawal et al. (2008) have demonstrated that MEL has significant anti-amnesic activity that may be attributed to increased cholinergic activity by inhibition of AChE activity in hippocampus. MEL has a multitude of other functions that could support most of the AD hypotheses such as Aβ accumulation, cholinergic, neuroinflammation, calcium, insulin resistance, and peroxidation of brain lipids hypothesis (Rosales-Corral et al., 2011). Thus, MEL is a potentially useful agent in attenuating and/or delaying AD.

In conclusion, this study demonstrates that EGT administration significantly improves memory and learning abilities in mice treated with Aβ1–40. This effect of EGT is likely through inhibition of Aβ accumulation in the hippocampus, which in turn significantly attenuates lipid peroxidation and protein damage, restores acetylcholinesterase (AChE) activity, and maintains GSH/GSSG ratio and SOD activity in brain tissues of Aβ1–40-induced mice. Although further studies are required, the present findings along with the fact that EGT has a transporter (OCTN1) in the neuronal cells to facilitate its entrance to the central nervous system suggest that EGT has the potential to become a useful neuroprotective agent for AD.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Acknowledgement

This study was supported by a Grant from the National Science Council, ROC (NSC 97-2313-B-235 -001 -MY3).

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